

MongoDB Installation, Configuration and CRUD Operations

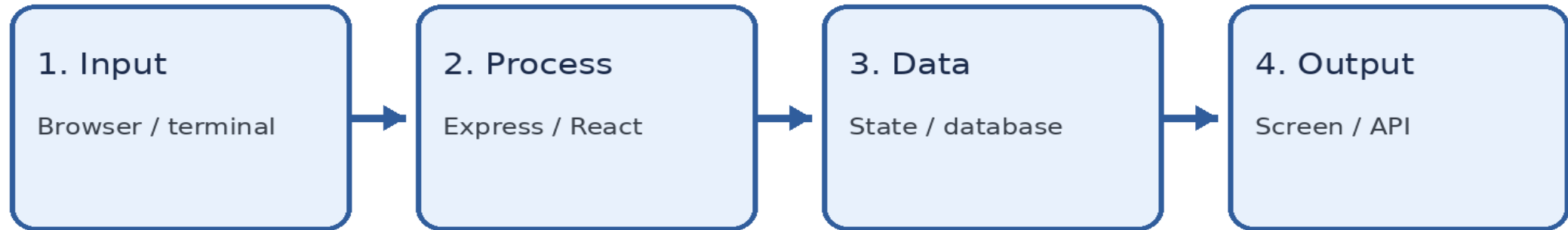
GitHub-readable preview of the presentation. Download the matching .pptx for the editable PowerPoint.

1. Syllabus tasks

- Install MongoDB and configure Atlas.
- Perform CRUD operations using `insert()`, `find()`, `update()` and `remove()`.

2. Workflow

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Workflow: input → processing → data/state → output

3.1. Code pathway — queries.js

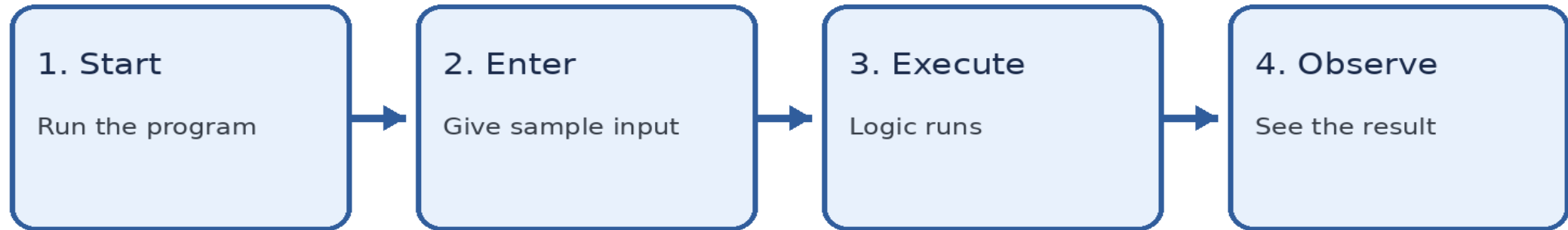
```
1 // FSD2 MongoDB CRUD lab. Run this file in mongosh.
2 use fsd2lab;
3
4 // INSERT: syllabus uses insert(); modern MongoDB also supports insertOne().
5 db.students.insertOne({ name: 'Rama', branch: 'CSE', year: 3 });
6 db.students.insertOne({ name: 'Gandhi', branch: 'CSE', year: 3 });
7
8 // FIND records.
9 db.students.find();
10 db.students.find({ branch: 'CSE' });
```

3.2. Code pathway — queries.js

```
11
12 // UPDATE: syllabus says update(); modern form is updateOne().
13 db.students.updateOne({ name: 'Rama' }, { $set: { year: 4 } });
14
15 // REMOVE: syllabus says remove(); modern form is deleteOne().
16 db.students.deleteOne({ name: 'Gandhi' });
17
18 // Check the final data.
19 db.students.find();
```

4. Runtime walkthrough

Runtime walkthrough — MongoDB Installation, Configuration and CRUD Ope



Runtime: start → enter → execute → observe

5. Run and observe

- Start the server / Vite app or open the MongoDB shell as described in the experiment README.
- Enter the sample input or call the required route.
- Watch the terminal for logs and errors.
- Verify the browser, API response, or database result.

6. Viva questions

- What is the purpose of this experiment?
- Which input is required?
- Which line or command performs the main operation?
- What output should you observe?
- What happens when the input changes?